Intermediate Elective

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Packages

```r
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(lubridate)
library(dplyr)
library(glue)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

```r
extrafont::loadfonts()
```

Data

```r
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

```r
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

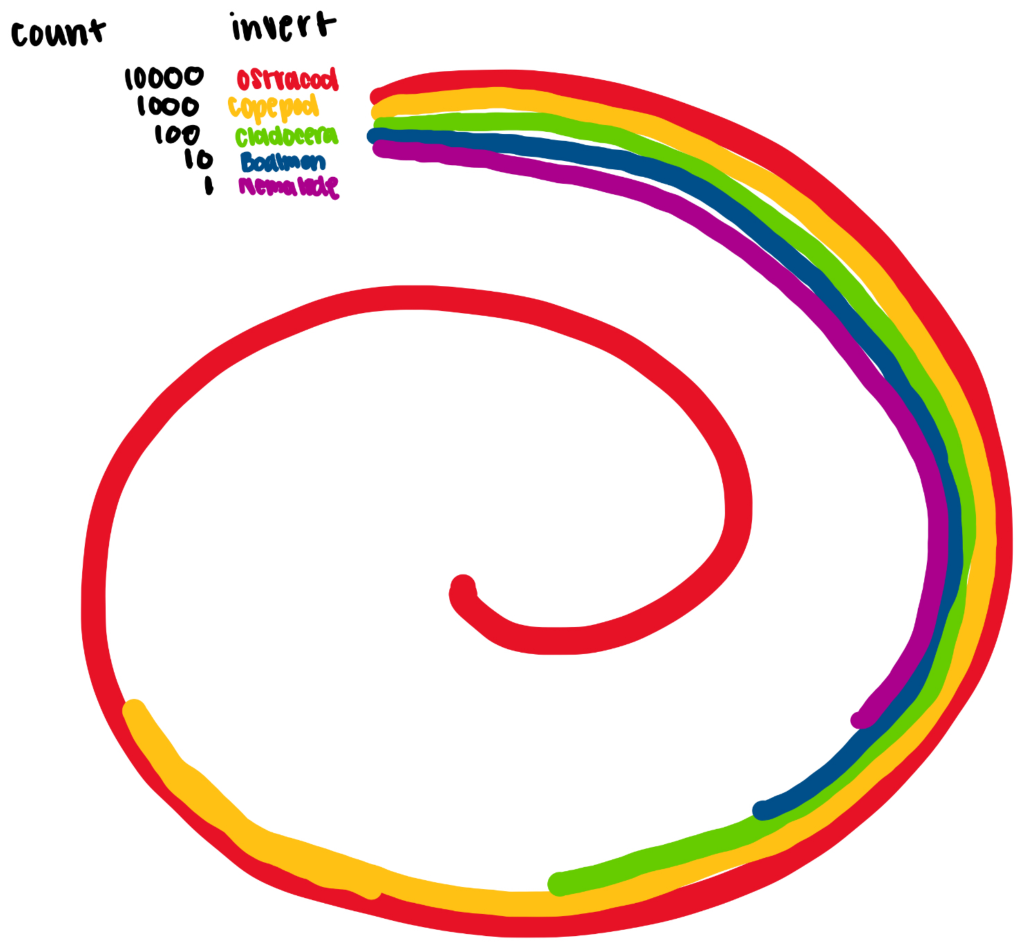
1. Explain your inspiration

- I picked the spiral looking graphic labled 2021_06_dubois_data graphic by Ijeamaka Anyenes.
- This graphic makes sense for my dataset of choice because it is able to display differences in abundance in a fun yet clear and simple way.
- The visualization will show the 5 most abundant inverts during 2024. Each line will represent a different invert and have its own color while the length of the line represents abundance.

2. Plan your figure

- Final visual goal: compare the abundances of the most common inverts found at ncos in 2024

```r
knitr:: include_graphics(here("code","IMG_2970.jpg"))
```



#3. Code your figure

Create data frame

```r
# create new data frame
invert_2024 <- aq_ins |>
  clean_names() |>
  rename(date = date_on_vial) |>
  # only include 5 listed invert species sampled in 2024
```

```r
  filter(year(date) == 2024) |>
  select(ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode) |>
  # change names of columns
  pivot_longer(cols = ostracod:nematode,
               names_to = "taxon",
               values_to = "abundance") |>
  # replace underscores with spaces
  mutate(
    abundance = replace_na(abundance, 0),
    taxon = str_to_title(str_replace_all(taxon, "_", " "))
  ) |>
  # group by taxon
  group_by(taxon) |>
  # add up total abundance for each taxon
  summarise(total_abundance = sum(abundance), .groups = "drop") |>
  # aggrange taxa from highest to lowest abundance
  arrange(total_abundance)
```

```r
# find max x value for the end of the spiral and divid by 1.5 to make the
# spiral wrap tighter
max_x <- max(invert_2024$total_abundance) / 1.5
# calculate the slope to create a spiral shape
slope <- (2-0) / (0 - max_x)
```

```r
# create new data frame for spiral segments
invert_spiral <- invert_2024 |>
  # give each taxon a different starting y position and separate segments
  mutate(
    taxon = as.factor(taxon),
    y = seq(12, by = -3, length.out = n()),
    # start all segments at x = 0
    x = 0
  ) |>
  # go one taxon at a time
  rowwise() |>
  # the segments either go to taxon abundance or stop at x max
  mutate(
    xend = min(total_abundance, max_x),
```

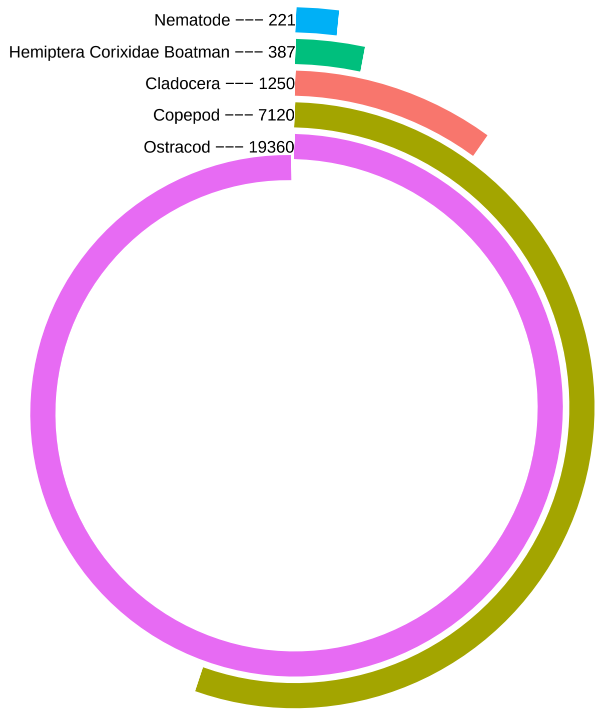
```
    # use slope find the ending y position
    yend = slope * xend + y
  ) |>
  # start second second where the first one ends so it wraps around
  mutate(
    y_2 = yend,
    # if abundance is larger than max x create a second segment
    x_2 = 0,
    xend_2 = if_else(total_abundance <= max_x,
                     NA_real_,
                     total_abundance - max_x),
    # calculate ending y position of second segment
    yend_2 = slope * xend_2 + y_2
  )
```

```
invert_label <- invert_spiral |>
  select(taxon, total_abundance, y, x) |>
  mutate(
    label = glue("{taxon} --- {total_abundance}")
  )
```

```
ggplot(data = invert_spiral) +
  geom_segment(aes(x = x, xend = xend,
                   y = y, yend = yend,
                   color = taxon),
               size = 5) +
  geom_text(data = invert_label,
            aes(x = x, y = y,
                label = label),
            size = 2.5,
            hjust = 1) +
  labs(
    title = "Total abundance of the most common inverts at NCOS in 2024",
    caption = "Source: 2024 invertebrate abundance data"
  ) +
  coord_polar(clip = "off") +
  ylim(-25, 15) +
  xlim(-25, max(invert_2024$total_abundance)/1.5) +
  theme_void() +
  theme(
    plot.background = element_rect(fill = "white",
                                   color = NA),
```

```r
    plot.margin = margin(t = 20, r = 5, b = 5, l = 5),
    panel.background = element_rect(fill = "white",
                                    color = NA),
    plot.title = element_text(hjust = 0.5,
                              face = "bold",
                              size = 15),
    plot.caption = element_text(size = 6),
    legend.position = "none"
  )
```

Total abundance of the most common inverts at NCOS in 2024



Source: 2024 invertebrate abundance data